

Carbohydrate Chemistry

Definition, Classification, Examples and Blood Glucose Regulation

Quick exam idea: Carbohydrates are the chief dietary source of energy, and blood glucose is tightly maintained by the opposing actions of insulin and hyperglycemic hormones.

1. Definition of Carbohydrates

Carbohydrates are organic compounds that are chemically defined as polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis.

They are mainly composed of carbon, hydrogen and oxygen. The older idea that all carbohydrates are "hydrates of carbon" is not fully correct, because some true carbohydrates do not follow the general formula exactly.

The term sugar is usually applied to carbohydrates that are sweet in taste, crystalline and soluble in water.

2. Important Functions of Carbohydrates

Function	Explanation / Example
Energy source	They are the most abundant dietary source of energy and provide about 4 kcal/g.
Energy storage	Glycogen stores glucose in animals and helps meet immediate energy needs.
Structural role	Cellulose in plants, glycosaminoglycans in connective tissue and carbohydrates of cell walls provide structural support.
Cell recognition	Glycoproteins and glycolipids on cell membranes help in cell adhesion, immunity, fertilization and receptor function.
Precursors	Carbohydrate metabolism supplies intermediates for synthesis of fats, amino acids and other biomolecules.

3. Classification of Carbohydrates

Carbohydrates are broadly classified according to the number of sugar units obtained on hydrolysis: monosaccharides, oligosaccharides and polysaccharides.

Classification of Carbohydrates

Based mainly on the number of sugar units liberated on hydrolysis

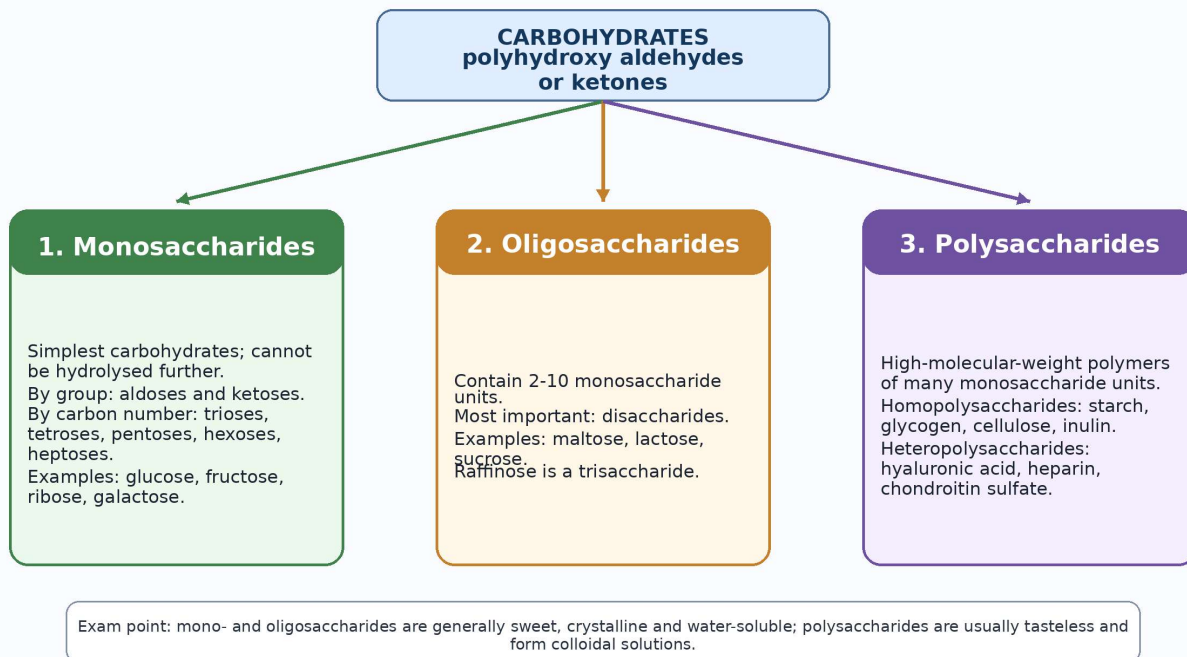


Fig. 1. Clean classification diagram of carbohydrates.

A. Monosaccharides

Monosaccharides are the simplest carbohydrates. They cannot be hydrolysed further into smaller carbohydrates. They may be classified by functional group and by number of carbon atoms.

Basis	Class	Meaning	Examples
Functional group	Aldoses	Contain an aldehyde group	Glyceraldehyde, glucose
Functional group	Ketoses	Contain a keto group	Dihydroxyacetone, fructose
Carbon number	Trioses	3 carbon atoms	Glyceraldehyde, dihydroxyacetone
Carbon number	Pentoses	5 carbon atoms	Ribose, ribulose, deoxyribose
Carbon number	Hexoses	6 carbon atoms	Glucose, fructose, galactose
Carbon number	Heptoses	7 carbon atoms	Sedoheptulose, glucoheptose

B. Oligosaccharides

Oligosaccharides contain 2-10 monosaccharide units. Disaccharides are the most important oligosaccharides in basic biochemistry.

Oligosaccharide	Composition	Type	Important point
Maltose	Glucose + glucose	Reducing disaccharide	Product of starch digestion
Lactose	Glucose + galactose	Reducing disaccharide	Milk sugar
Sucrose	Glucose + fructose	Non-reducing disaccharide	Cane sugar / table sugar
Raffinose	Galactose + glucose + fructose	Trisaccharide	Found in plants and legumes

C. Polysaccharides

Polysaccharides are high-molecular-weight polymers of monosaccharides joined by glycosidic bonds. They are usually tasteless and form colloidal solutions with water.

Type	Definition	Examples	Importance
Homopolysaccharides	Made of one type of monosaccharide	Starch, glycogen, cellulose, inulin	Storage or structural function
Heteropolysaccharides	Made of more than one type of monosaccharide or derivatives	Hyaluronic acid, heparin, chondroitin sulfate	Extracellular matrix, anticoagulation, tissue support

Memory line: Starch = plant storage, glycogen = animal storage, cellulose = plant structural polysaccharide, inulin = used to assess glomerular filtration rate.

4. Important Examples and Biological Importance

Carbohydrate	Class	Biological importance
Glucose	Aldohexose monosaccharide	Main blood sugar and major immediate fuel for tissues.
Fructose	Ketohexose monosaccharide	Fruit sugar; converted in liver into metabolic intermediates.
Galactose	Aldohexose monosaccharide	Component of lactose and glycolipids.
Ribose	Pentose monosaccharide	Present in RNA and nucleotides such as ATP.
Deoxyribose	Pentose derivative	Sugar component of DNA.
Maltose	Disaccharide	Intermediate product in starch digestion.
Lactose	Disaccharide	Milk sugar; made of glucose and galactose.

Carbohydrate	Class	Biological importance
Sucrose	Disaccharide	Common dietary sugar; made of glucose and fructose.
Starch	Homopolysaccharide	Major storage carbohydrate in plants.
Glycogen	Homopolysaccharide	Storage carbohydrate of animals; abundant in liver and muscle.
Cellulose	Homopolysaccharide	Structural carbohydrate of plants; dietary fiber in humans.
Heparin	Heteropolysaccharide	Natural anticoagulant.

5. Blood Glucose Regulation

Blood glucose regulation means maintaining the glucose concentration in blood within a narrow range despite feeding, fasting, exercise and stress.

In a healthy adult, fasting blood glucose in the postabsorptive state is about 70-100 mg/dL. After a carbohydrate meal, it may rise to about 120-140 mg/dL. Plasma glucose values are slightly higher than whole-blood glucose values.

Blood glucose level depends on two opposing processes: glucose entering the blood and glucose being removed or utilized by tissues.

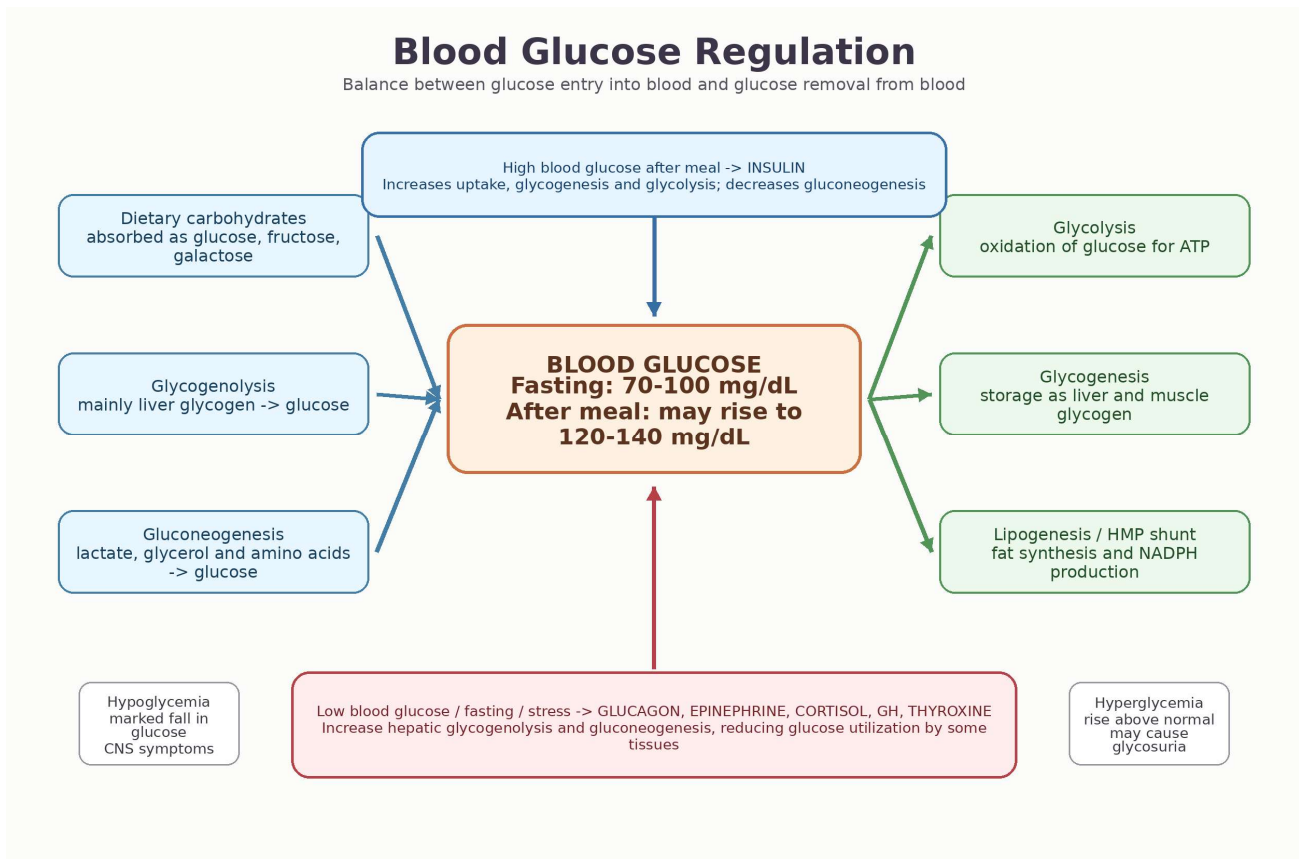


Fig. 2. Readable overview of blood glucose homeostasis.

A. Sources of blood glucose

- Dietary carbohydrates are digested and absorbed as monosaccharides. Glucose enters the blood directly, while fructose and galactose are mainly converted in the liver.
- Glycogenolysis in the liver breaks stored glycogen into glucose, especially during early fasting.
- Gluconeogenesis forms glucose from non-carbohydrate precursors such as lactate, glycerol and glucogenic amino acids, mainly during fasting.

B. Utilization/removal of blood glucose

- Glycolysis oxidizes glucose to provide ATP.
- Glycogenesis stores glucose as glycogen in liver and muscle.
- Excess glucose can be converted to fatty acids and triacylglycerols.
- Some glucose enters the hexose monophosphate shunt to produce NADPH and ribose sugar.

C. Hormonal control

Hormone	When active	Main action on blood glucose
Insulin	After meals / high blood glucose	Lowers blood glucose by increasing glucose uptake, glycogenesis and glycolysis, and by decreasing gluconeogenesis and glycogenolysis.
Glucagon	Fasting / low blood glucose	Raises blood glucose by stimulating hepatic glycogenolysis and gluconeogenesis.
Epinephrine	Stress and exercise	Raises blood glucose by stimulating glycogen breakdown, especially during acute stress.
Glucocorticoids	Prolonged stress / fasting	Raise blood glucose by increasing gluconeogenesis and decreasing glucose utilization in some extrahepatic tissues.
Growth hormone and ACTH	Hormonal/stress responses	Tend to increase blood glucose by reducing glucose utilization.
Thyroxine	Thyroid hormone effect	Can elevate blood glucose by stimulating hepatic glycogenolysis and gluconeogenesis.

D. Fed state vs fasting state

- Fed state: insulin predominates; glucose enters tissues and is stored as glycogen or converted to fat.
- Early fasting: glucagon predominates; liver glycogenolysis maintains blood glucose.
- Prolonged fasting: gluconeogenesis becomes more important; lactate, glycerol and glucogenic amino acids are converted to glucose.

6. Clinical Terms to Remember

Term	Meaning
Hyperglycemia	Blood glucose rises above normal; persistent hyperglycemia is important in diabetes mellitus.
Hypoglycemia	Marked decrease in blood glucose. Severe hypoglycemia affects the brain and may cause headache, anxiety, confusion, sweating, seizures or coma.
Glycosuria	Excretion of glucose in urine, commonly seen when blood glucose exceeds the renal threshold.

Final revision box: Insulin is the major hypoglycemic hormone. Glucagon, epinephrine, glucocorticoids, growth hormone, ACTH and thyroxine are hyperglycemic hormones.

7. One-page Recall Summary

- Definition: polyhydroxy aldehydes or ketones, or compounds that yield them on hydrolysis.
- Main classes: monosaccharides, oligosaccharides and polysaccharides.
- Monosaccharide examples: glucose, fructose, galactose, ribose and deoxyribose.
- Disaccharide examples: maltose, lactose and sucrose.
- Polysaccharide examples: starch, glycogen, cellulose, inulin, heparin and hyaluronic acid.
- Blood glucose is maintained by balancing sources of glucose with tissue utilization and storage.
- Insulin decreases blood glucose; glucagon and other stress hormones increase blood glucose.